Q.1) State Inclusion / Exclusion principal with Proof.

Sol^n –

The Inclusion - Exclusion principle is an important rule in set theory, combinations, and Probabity. it is used to find the total number of elements in the union of two or more sets without counting the common elements more than once. when sets have overlapping elements direct addition of their sizes. causes overcounting. The Inclusion-Exclusion Principle helps to correct this by adding and subtracting the intersections of sets in a systemetic way.

Statement :

For two sets A and B, the Formula is,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Here, $n(A)$ and $n(B)$ represent the number of elements in each set, and $n(A \cap B)$ represents the number of elements common to both.

When $n(A)$ and $n(B)$ are added, the common elements are counted twice. so, substracting $n(A \cap B)$ once give the correct total.

For three sets A, B and C :

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - [n(A \cap B) + n(B \cap C) + n(C \cap A) + n(A \cap B \cap C)]$$

Here, we first add all sets, then substract pairwise intersections to remove double counting and finally add the triple intersections because it was substracted too many times.

Proof (For Two sets) :
When we add $n(A) + n(B)$, elements in both A and B are counted twice - once in A and once in B. To count them only once, we substract the number of elements in $A \cap B$
Thus,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Ex: let $n(A) = 40$, $n(B) = 30$ and $n(A \cap B) = 10$
Then, using the ~~function~~, formula :

$$n(A \cup B) = 40 + 30 - 10 = 60$$

So, there are 60 distinct element in the union of sets A and B.

The Inclusion-Exclusion Principle ensures that each element in combined sets is counted exactly once. It is widely used in counting problems, Probability theory, and logical analysis, especially when dealing with overlapping data or multiple condition

Q.2) Define Well-Formed Formula and write its rule

Soln —

A well-formed formula is a fundamental concept in propositional logic and predicate logic, it refers to a formula that is syntactically correct, meaning it follows all the grammatical and structural rules of logical expression. In simple terms, a WFF is a meaningful and valid statement formed by combining propositional variables and logical connectives. in the correct orders. every well-formed formula expresses a logical proposition that can be evaluated as either true or false, but not ~~to~~ both.

Definition:

A Well-Formed Formula is a valid expression built from propositional variables (such as P, Q, R, S, ---) and logical operators (such as $\sim$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$) by following specific formation rules. The purpose of these rules is to ensure that the logical expression is meaningful and interpretable within the system of logic

For example:

- $P \rightarrow$ WFF (a single proposition)
- $\sim P \rightarrow$ WFF (negation of a proposition)
- $(P \wedge Q) \rightarrow$ WFF (combination of two Propositions)

- Rules for forming WFF {in

Rule 01 : Atomic formulas are WFFs every propositional variable such as

- P, Q, R, A_1, B_2, etc is a WFF

Rule 02 : If P is a WFF, then ~P is also a WFF.

ex : If P is a WFF → ~P is also a WFF.

Rule 03 : If P and Q are WFFs then the following are also WFFs.

using binary connectives –

- $(P \land Q)$ – AND
- $(P \lor Q)$ – OR
- $(P \rightarrow Q)$ – implication
- $(P \leftrightarrow Q)$ – Biconditional

Rule 04 : Parentheses must be used properly. Parentheses ensure the correct grouping of expressions.

ex : $(P \lor Q) \land R$ is not the same as $P \lor (Q \land$

Rule 05 : Only expressions formed using the above rules are WFFs. anything that violates these rules is not a WWF.

incorrect ex :
- $PQ\land$ {missing operator}
- $\land PQ$ {operator misplaced}
- $P\lor$ {incomplete}
- $P(Q \rightarrow)$ {incorrect structur

Q.3) Proof : Every non-empty set of positive integers has a least element.

Soln —

Every non empty set of positive integers has a least elements

this is known as the well-ordering principles or well-ordering property of positive integers. it is one of the fundamental properties of the set of natural numbers $N = \{1, 2, 3, ---\}$

To Proove :
IF S is a non-empty set of positive integers, then S contains a least element - that is, there exists an element $m \in S$ such that

$m \leq x$ for all $x \in S$

• Proof (By condradiction) :

(1) let S be a non-empty set of positive integers
(2) Suppose, for the sake of contradication, that S has no least element.
(3) this means : For every element $x \in S$, there exists another elements $y \in S$ such that $y < x$.

(3) Since S is non-empty, it must contain at least one positive integer, say $a \in S$

(4) Now by assumption, there must be another element $b \in S$ such that $b < a$. but since a is a positive integer b must also be a positive integer and $b < a$.

(5) Repeating this process, we can continue finding smaller positive integers in S.

(6) However, the positive integers ~~control~~ cannot decrease indefinitely, because these is no positive integer smaller than 1. the smallest positive ~~is 1~~ integer is 1.

(7) This leads to a contradiction, because if $1 \in S$, then 1 would be the least element. if $1 \notin S$, we cannot have an infinite decreasing sequence of Positive integer - the process must stop at some smallest integer.

(8) Therefore, our assumption that S has no least element is False.

- Hence, every non-empty set of positive integer has a least element:

This properly is called the well-ordering principle and it is fundamental to many proofs in number theory and mathematical induction.

Q.4) Show that $1^2 + 2^2 + 3^2 + ---- + n^2 = n/6 (n+1)(2n+1)$, $n > 1$ by mathematical induction.

Sol^n —

$$1^2 + 2^2 + 3^2 + ----- + n^2 = \frac{n(n+1)(2n+1)}{6}$$

for all integers $n \geq 1$.

Proof (by mathematical induction)

Base case $(n=1)$

left side $= 1^2 = 1$

Right side $= \frac{1\cdot(1+1)\cdot(2\cdot1+1)}{6} = \frac{1\cdot2\cdot3}{6} = 1$

So the formula holds for $n = 1$.

Induction Hypothesis :

Assume the formula holds for some $k \geq 1$

$$1^2 + 2^2 + ---- k^2 = \frac{k(k+1)(2k+1)}{6}$$

Induction step :

consider the sum up to $k+1$:

$$1^2 + 2^2 + --- + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$

(using the induction hypothesis) :

factor out $(k+1)$:

$$= \frac{(k+1)(k(2k+1) + 6(k+1))}{6} =$$

$$\frac{(k+1)(2k^2 + k + 6k + 6)}{6} = \frac{(k+2)(2k+2)}{6}$$

Now factor the quadratic :
$2k^2 + 7k + 6 = (k+2)(2k+3)$. Thus

$$1^2 + \dots + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

But the right hand side is exactly the formula with $n = k+1$:

$$\frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$$

since the statement is true for $n = 1$ and the truth for $n = k$ implies the truth for $n = k+1$, it follows by induction that the formula holds for all integer $n \geq 1$ (and therefore for all $n > 1$).

Q.5) Construct the Truth Table for $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \wedge r$ and state whether it is equivalent or not.

Sol^n —

$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \wedge r$

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	$p \wedge q$	$(p \wedge q) \wedge r$
T	T	T	T	T	T	T
T	T	F	T	T	T	F
T	F	T	T	T	F	F
T	F	F	F	F	F	F
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	F	F	F
F	F	F	F	F	F	F

Since the truth values of $p \wedge (Q \vee R)$ and $(p \wedge q) \wedge r$ are not identical for all possible combinations of p, q and r the two statements are not logically equivalent.

Scanned with OKEN Scanner